Printed Circuit boards

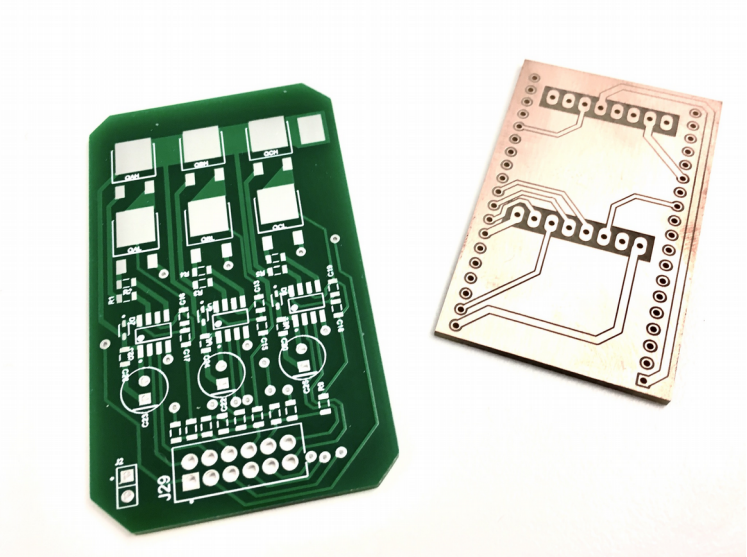
About

This text is intended to give a brief introduction of important concepts that are involved in the design of PCBs.

What is a PCB?

To avoid confusion lets start with defining what we mean when we talk about a printed circuit board (PCB). A PCB could be anything from a “veroboard” made from phenolic paper, a protoboard, or a professionally manufactured 6 layer card for hundreds or even thousands of components.

What these things have in common is that they have at least one layer of a non conductive material (called a substrate) and at least one layer of electrically conductive tracks.



When do we need a PCB?

There are many reasons why you might want to use a PCB. For early prototypes you can often get away with using break-out boards, development boards, through hole components and a breadboard. However there are some circumstances when that will not be sufficient. Below are some situations when that might be the case.

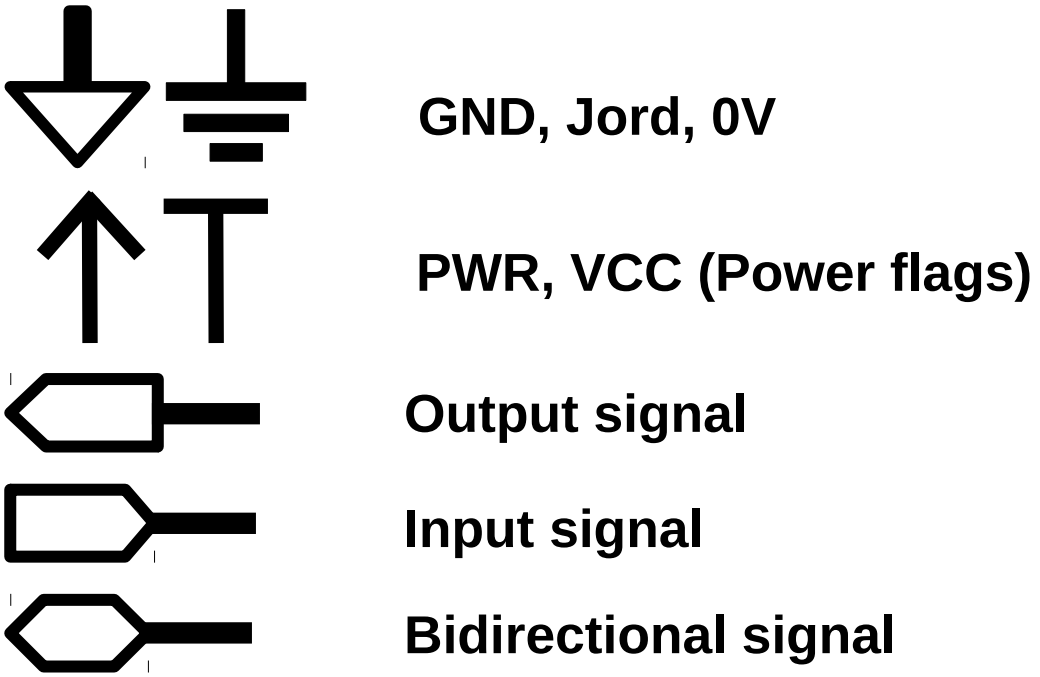
- **Complexity** – When your circuit starts to approach a couple hundred connections a breadboard design can start to become hard to maintain. The time you spend looking for bad connections and misplaced components can quite quickly add up to a lot more than the time spent designing a PCB for the project.
- **Availability –** Increasingly components have stopped being made in a breadboard friendly package. Many components aren't even released in a hand solderable package today. Sometimes components like this will have a breadboard friendly development board available. But if you want to use an obscure part which doesn't have one available going for a custom PCB will often be the best option.
- **Pin count –** This goes in almost the same category, but sometimes a component you want to use just has too many pins to be practically usable on a breadboard. A typical large breadboard usually has room for around 240 nodes, and there are plenty of IC's with a lot more pins than that.
- **Signal integrity** – A breadboard is usually quoted to be able to handle digital signals up to around 10MHz without degrading the signal too much. Sometimes 10MHz is not enough speed for your signals. A properly designed PCB can handle signals with frequency components exceeding 3GHz.
- **Size** – Some prototypes have size related requirements to be tested properly. If you are building a wearable sensor to be used in a sport it's not very practical if the prototype consists of two separate breadboards and a crows nest of wires. Making a PCB often enables the design to be shrunk considerably.
- **Production prototypes** – When designing a product meant for large scale production it is a good idea to make the design as easy as possible to manufacture. Designing a PCB in an early stage of the prototype allows more refinement towards the final production PCB. (Since hopefully the final product will not include a breadboard and a mess of wires)
- **Investor demo** – When looking to get an investment into your idea from an outside party a sleek looking PCB will always be more impressive than a breadboard prototype, regardless if the function is exactly the same.

Schematics

The first part of designing a PCB involves deciding which components to use and how they should be connected. This is done in the schematic part of your ECAD-tool (Electronic computer aided design) and should look familiar to you if you have made any circuits for simulation before.

Reading a schematic

Schematics use a number of basic symbols which you need to know about to be able to correctly read a schematic. These are some of the most common symbols.



The symbols in a schematic is used both to connect different parts of a design, places with the same symbol (and name) will be considered connected to the same node. The symbols are also used to communicate the intention of the designer. For example the input and output flags does not affect the PCB layout in any way but makes the signals function easier to interpret.

Schematics for physical design

When designing a schematic for a PCB there are some different considerations compared to when designing for simulation. Every input and output needs some form of connection point. Whether that is an external analog signal, a digital signal or a voltage/current supply.

In a real world design a 5V power supply can mean a multitude of things, a pack of batteries, a wall plug, a USB cord, solar cells etc. All of these different power supplies would use a different type of connector and therefore the design needs to reflect that in the form of an appropriate component.

Power supplies and voltage regulation

An electrical circuit will often require multiple regulated voltages. For example an instrumentation amplifier might require a positive and an negative supply. A microcontroller might require 3.3V while a connected sensor needs 1.8V to function.

This can be achieved in a few ways, but the most common would be to either have multiple connectors with different voltages. This is for example the approach used by ATX compatible motherboards, which has conductors for 12V, -12V, 5V, -5V, and 3.3V in its power connector.

The more common approach however is to take a single voltage through a connector and do the conversion directly on the PCB. Different ways to convert the voltage exists and will typically depend on the application. The table below describes the most common ways.

Method	Regulates up/down?	Pros	Cons
Voltage divider	Down	By far the cheapest solution, and has very low complexity.	Can only be used as a reference, since any load on it will change the voltage.
Low drop out regulator (LDO)	Down	Simple to use. Requires few components. Usually very low noise.	There's always a voltage drop from the supply. Not very efficient.
Switching buck converter	Down	Efficient. No inherent voltage drop required.	Somewhat more components needed to implement. Higher risk of ripple on the output.
Switching boost converter	Up	Efficient. Can regulate up.	Same as buck converter. Usually more expensive due to the inductor needed.
Switching buck/boost converter	Up/Down	Regulates in both directions. Very useful for input power supplies that varies around the output.	Same as buck+boost, and has the highest implementation complexity.

(The pros and cons here are quite generalized and implementation details can affect performance in either direction)

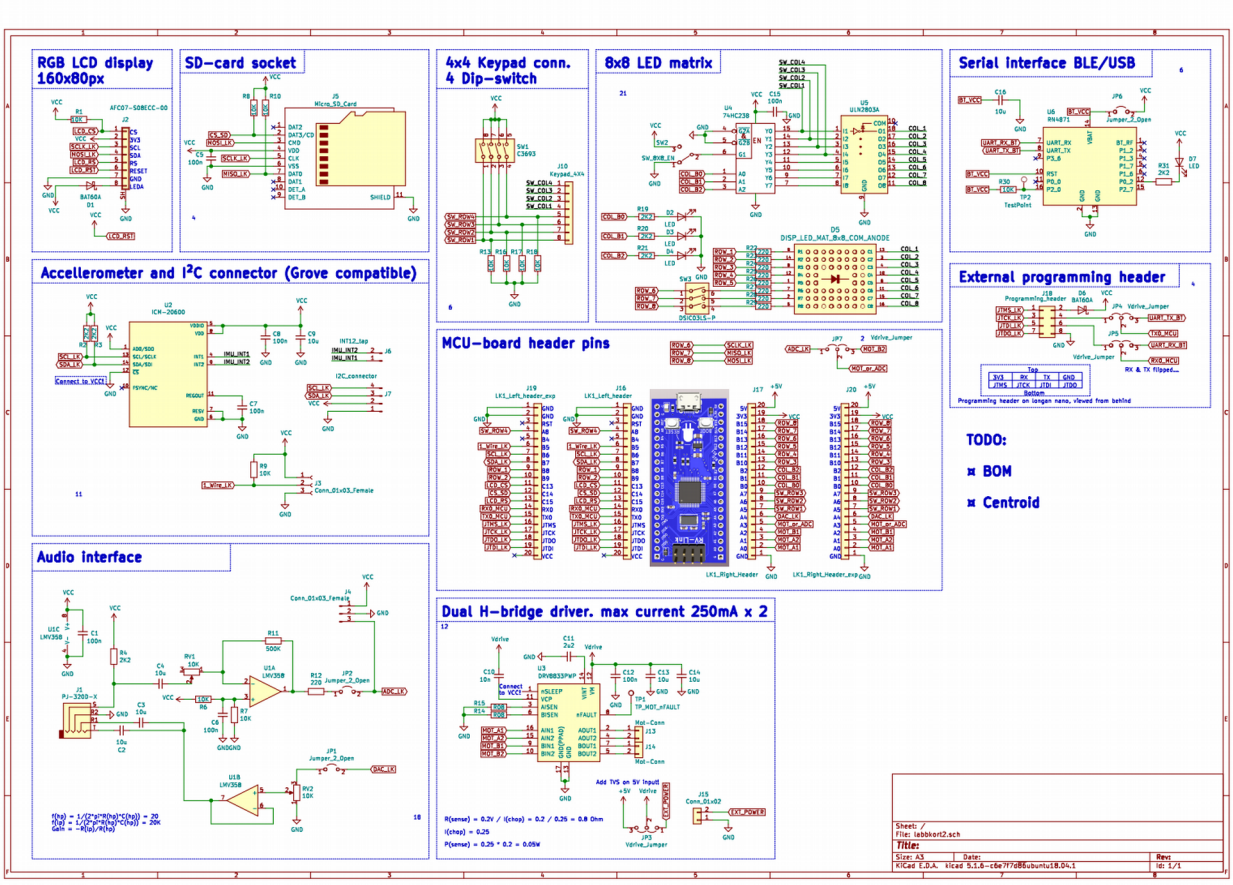
Modularity

Designing your schematic with modularity in mind is a good best practice for the same reason it's useful in other areas such as programming.

Modularity reduces unnecessary coupling between parts of the circuit and allows for a better overview of the function.

It also allows for easier reuse of modules in different designs.

Each module should be divided up with a single responsibility in mind. How abstract this responsibility should be depends on the design, but usually this division is a lot more natural for hardware than for software.



Example of a circuit divided into multiple modules. Although the components connect in the middle there is a clear division of the different responsibilities on the board.

Modules are connected by either using input/output flags, or by naming nets.

Inspiration

A good way to learn is to imitate what others have done. This also applies to circuit design. There is almost always someone else who has done something similar to what you are attempting.

In a lot of datasheets for components you can find complete schematics for how to implement the circuit. The manufacturers want you to use their component and the datasheet is what they produce to help you use it. Sometimes there will also be “app notes” associated with a component which can be incredibly helpful.

There is also a fairly new but lively open source community around hardware design called “open source hardware”. Designers involved here openly publish their designs, so you can see how a device has been implemented. Some notable sources for open source hardware below.

- Adafruit – Manufactures and sells a huge amount of different development boards and breakout boards, all open source.
- Olimex – Same as Adafruit. Designs all their designs in KiCad.
- OpenCompute – This is a great source if you have ever wondered how a schematic for a highly complex board might look. This project contains schematics for complex boards such as multi core server motherboards. Great learning resource for more advanced design.

Physical design

Designing a PCB also involves taking into account the shape and mounting style of the components.

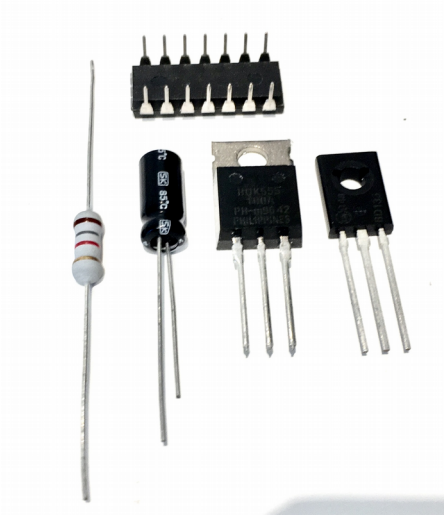
Electrical components for PCBs come in two major categories; SMT(Surface mount technology), and THT(Through hole technology).

The distinction between the two is the mounting style, where THT mounts in a hole through the board. The type of resistors used for breadboarding falls into this category for example.

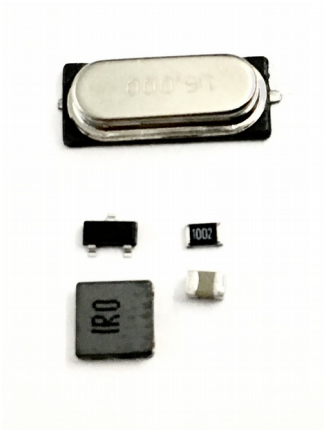
SMT components mounts by being placed on top of the PCB without conductors going through the card.

It is becoming more and more rare to see through hole components on boards meant for large scale production. This is due to a number of reasons, but the main factors are:

- Size, SMT components can be made a lot smaller than THT counterparts, resulting in smaller boards in general.
- Reduced costs for component placement, THT often cant easily be placed by a machine.
- Lowered board cost due to fewer drill operations.



Through hole (THT)

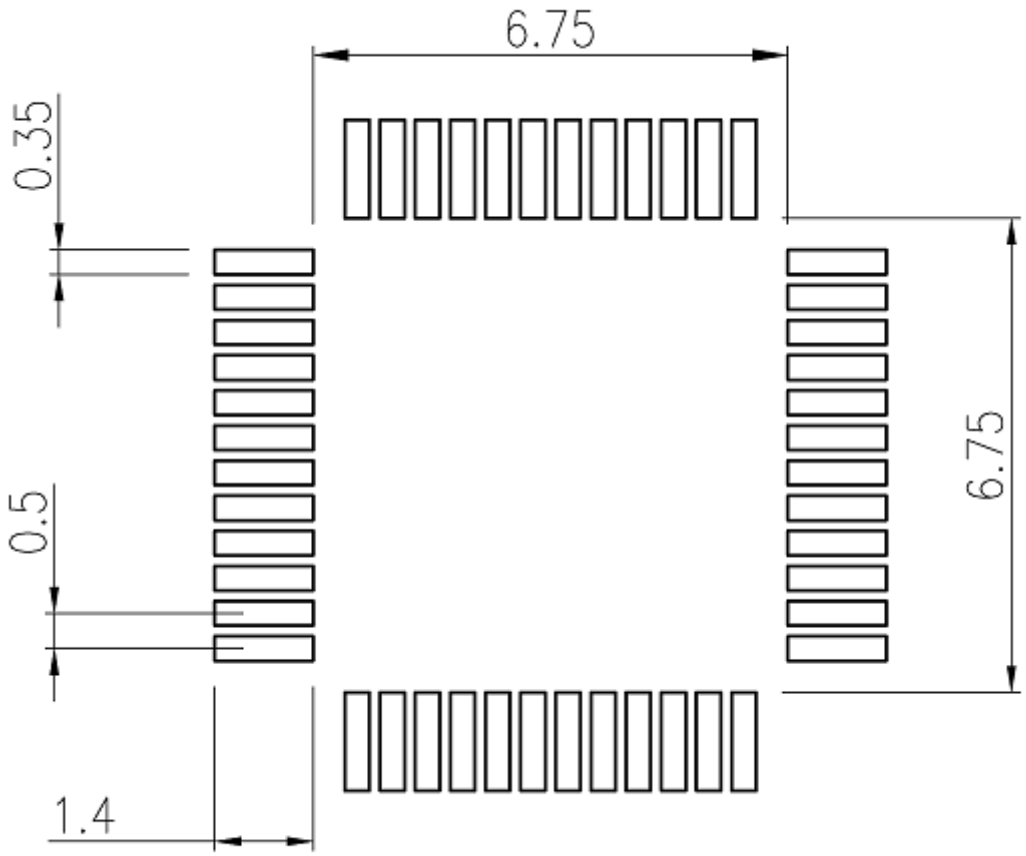


surface mount (SMD)

Footprint

For your ECAD program, the most important characteristic of your component isn't mainly its shape, rather it's the points of contact that the pins make to the board. This pattern is called the components "footprint".

The components footprint contains information of which pin connects to which pad, and the size and coordinates for each pad. It also contains some other informaton on size and shape of the soldermask, solder paste, and silkscreen.



Measurements for an LQFP48 footprint

Units

Due to some unfortunate history PCB design still has yet to be standardized on a single measurement system. This means that both the imperial and metric systems are used, often both on the same design.

Typically measurements will be in either millimeter or mils.

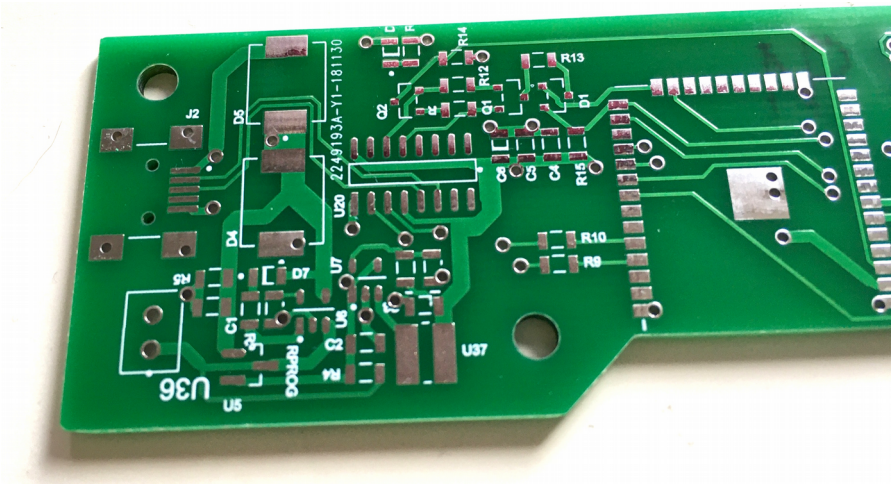
Mils also refered to as thous are a thousandth of an inch. Consequently a mil is equal to 0.0254mm and a millimeter is 39.37mils.

As a PCB designer you should be prepared to convert between the two more or less interchangably. Just make sure not to mix them up.

Layout

Once you have completed the schematic it’s time to start drawing the layout of the PCB. The layout will be a digital representation of the boards physical form. A typical PCB consists of a few different types of layers.

An ECAD program has one layer for each instance of each layer.



White – Silkscreen. Contains information on where components go and more.

Green – Solder mask, protects copper which should not be soldered

Paths under silkscreen - Copper

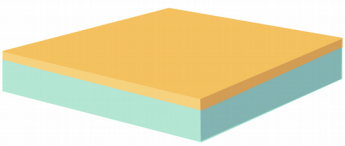
Substrate – The isolation layer, in this case FR4 (Glass fiber)

Copper layers

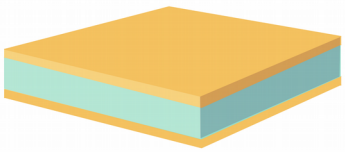
The two outermost layers of a PCB are referred to as “top copper” and “bottom copper”. A card can also have a number of inner layers.

For relatively simple designs a two layer board is usually sufficient and the most cost effective.

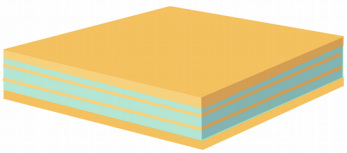
More complex designs will often require more layers than two but cost goes up with each added pair of layers.



Single layer



2-layer

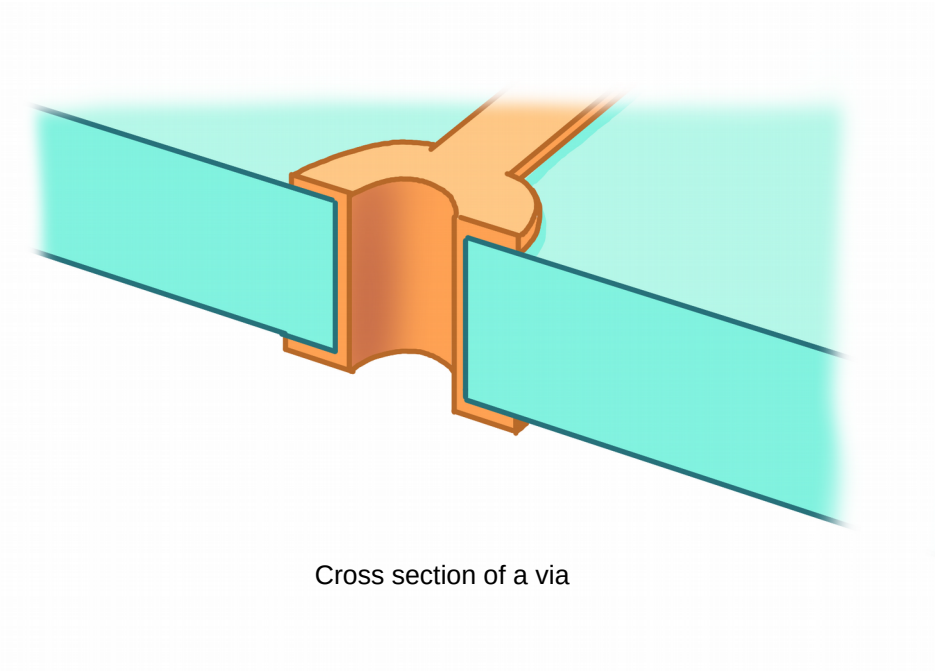


4-layer

The copper layers thickness can vary based on the current or thermal capability required for the board. The most common thicknesses are 1oz/ft² and 2oz/ft². 1oz/ft² of copper is equivalent to approximately 34,79µm.

Vias

To make a connection between two layers something called a via is used. A via is made by drilling a hole through the PCB and then connecting the two layers through the hole. In PCB factories this is usually done by electroplating the inside of the holes walls.



Cross section of a via

Soldermask and silkscreen

Soldermask is used to protect copper that should not be soldered. This is the layer which gives a typical PCB its green color. It is also possible to make boards with other colors on the soldermask. As an example, on the PCB below blue soldermask was used.



Silkscreen is used for annotating the PCB with directions for component placement and other useful information. On the PCB above the silkscreen layer are the white parts.

Routing

When going to the layout stage of the process each component in the design should have a corresponding footprint. How this association is made varies between different ECAD software.

At the layout stage the ECAD tool will place every component that is present in the schematic arbitrarily on the page. At this point the components will have no electrical connections to each other. There will however be lines indicating where a connection *should* go. The lines are called “Air wires” and together they are referred to as a “rats nest”.

Placement

The first thing you should do when making your layout is to place the components in a logical manner.

Two main factors contribute to the placement of components: Function, and connections.

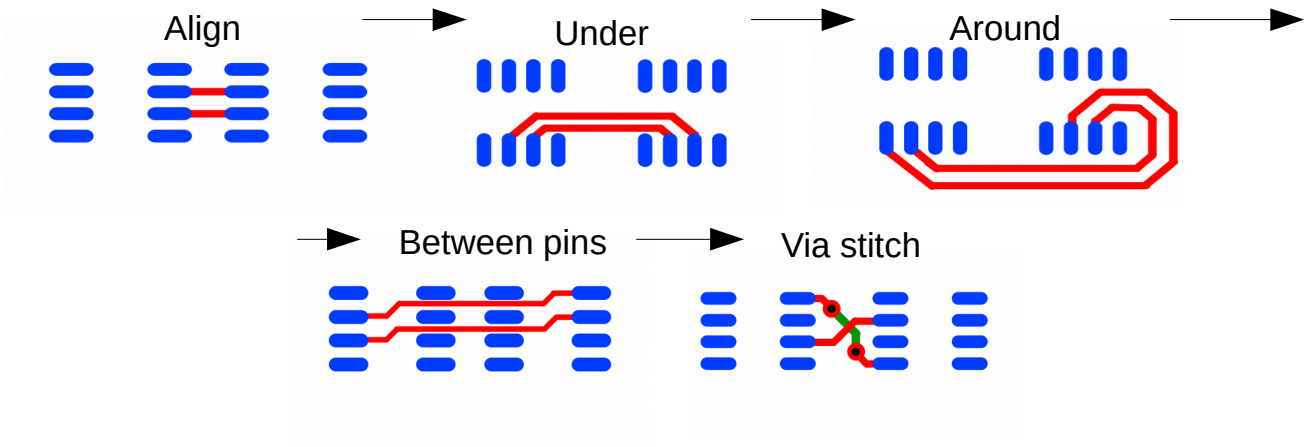
A functional placement might be that a temperature probe must be placed far away from a source of heat or an indicator LED that must be somewhere it can be seen.

Secondly the connections a component has also dictates where it is appropriate to put it. You want to place your components so that as few lines cross as possible. This makes routing the connections a lot easier.

Traces

After having placed the components in a good arrangement you can start to draw the traces connecting the different nodes. The traces are paths on the copper layer creating an electrical connection.

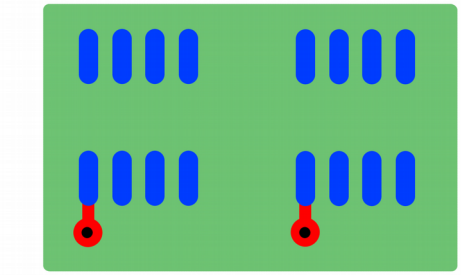
If no air wires cross you can usually just draw the trace directly. When there is a cross you need to use some strategy to get around it. Below are some ways that you can use in the order that I usually consider the layout.



Pours / Fills

In addition to traces you can also define areas to fill with copper, these areas are referred to as fills or pours.

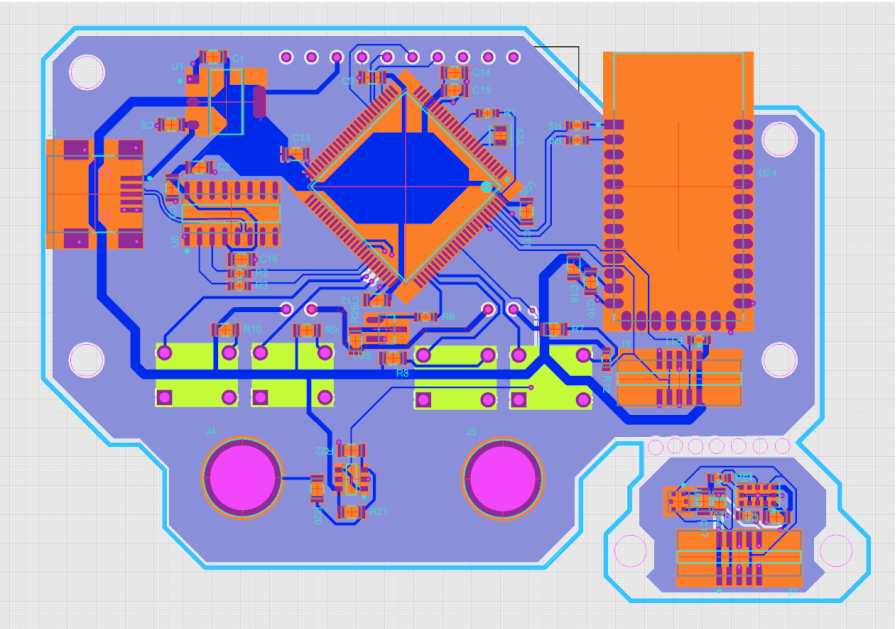
It is very common to use one layer (bottom on a 2-layer board) for a fill connected to GND. This simplifies routing since most components connect to GND, it also provides a very low impedance path for the ground, which can reduce overall noise.



Example pour where green is the bottom layer. It makes it very simple to connect both IC's to ground.

Outline / Edge Cuts

For a complete layout you also need to consider the shape of the board. This is defined as the boards outline. (Cyan in the picture below) Typically this information is on the layer "Mechanical layer" or an "Edge cuts" layer.



Non Ideal Properties of Copper Traces

Something that you often disregard when doing electrical simulations are the non-ideal properties of the conductors. In a lot of cases this can be mostly disregarded when making a PCB design as well.

There are however cases when the trace impedance does matter quite a bit.

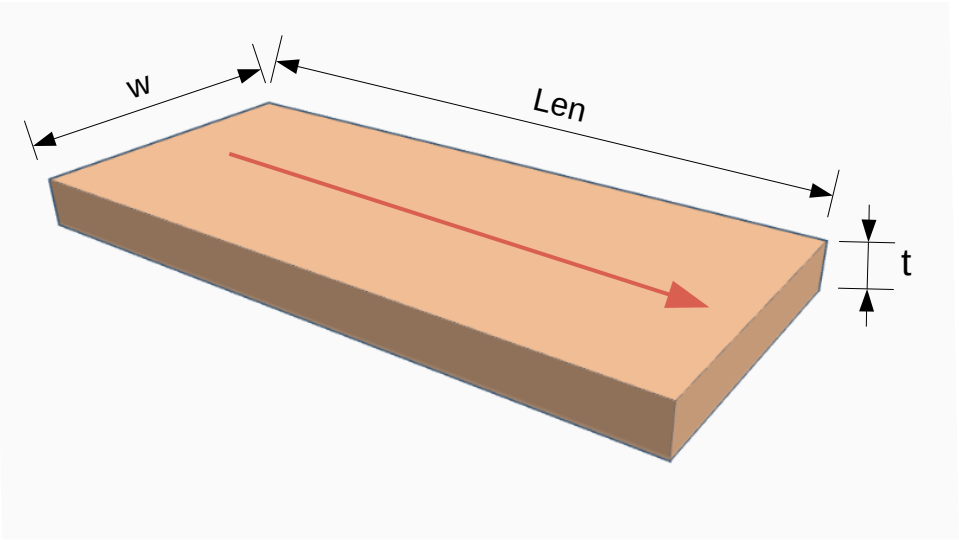
Copper resistance can cause heat, and if not accounted for correctly sometimes it can cause a circuit to fail.

You can quite easily calculate the resistance of a of a trace. The following formula is used used to calculate the resistance of a cuboid.

$$R = \rho \frac{Len}{w \, x \, t}$$

ρ = the materials permitivity

1.73μΩcm for copper



Since a traces thickness is constant ***t*** is always the same.

To get a useful characteristic we can calculate the resistance of any size square (Len = w) which gives the following formula: $R = \frac{\rho}{t}$

For a 1oz/ft² copper layer we get the following calculation:

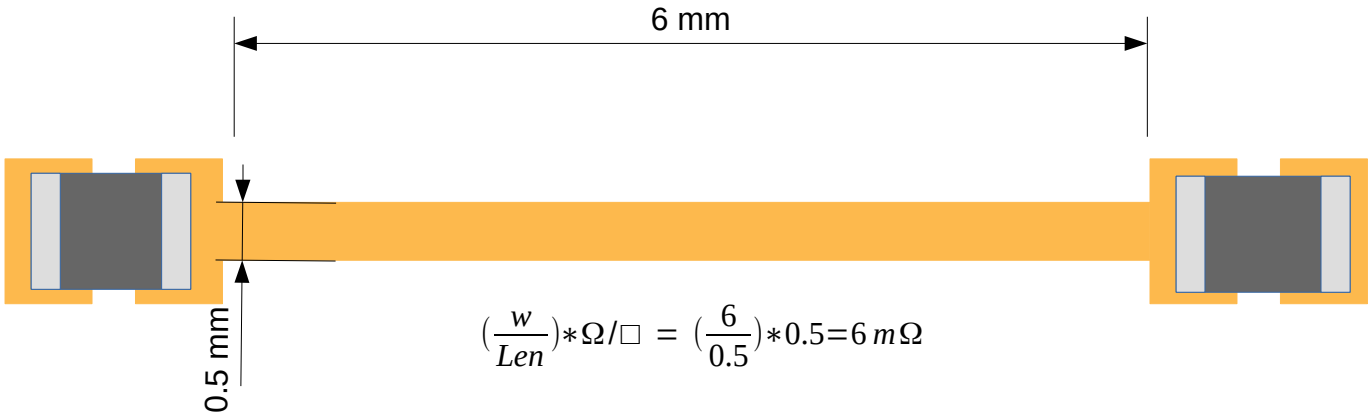
$$\rho_{cu} = 1.72 \mu \Omega cm$$

$$t_{1oz} = 34.7 \mu m$$

$$R = \frac{0.0172 \, \Omega}{34.7} = 0.5 \, m\Omega$$

This means that any size square of 1oz/ft² will have a resistance of 0.5mΩ whether its 1x1m or 1x1mm.

Using this characteristic we can just use the aspect ratio of the trace to get its resistance. For example a 0.5mm thick and 6mm long trace will have a resistance of 6mΩ, since it should have the same resistance as 12 squares placed in a row.



To know that a conductor has a low enough resistance is important when you need to conduct high currents.

When current passes through a resistive element heat is generated, and a voltage drop is present over it. For example, a 30mm long and 0,3mm wide copper conductor (1oz/ft² thick) will have a resistance of 50mΩ. If we pass 5A through this conductor it would have a voltage drop of 250mV and a heat loss of 1,2W. This would be more than enough to heat the conductor beyond the point of failure.

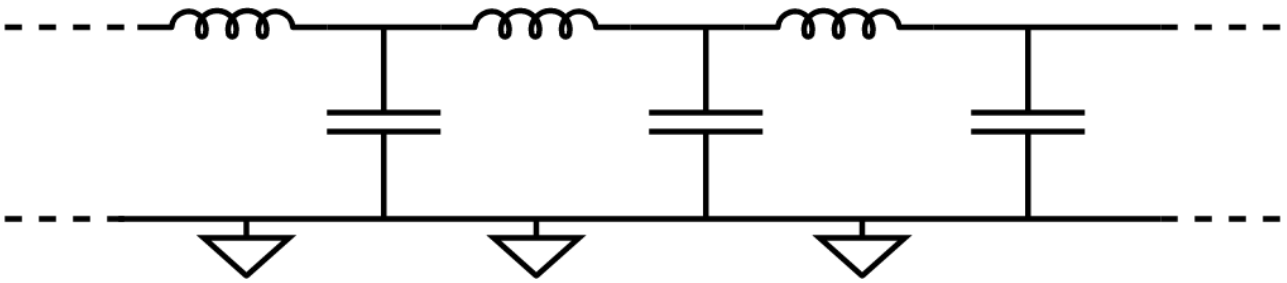
For an easy way to get correct trace widths for high current circuits you can use the handy online calculator below.

https://www.4pcb.com/trace-width-calculator.html

Trace impedance

At higher frequencies the capacitance and inductance becomes more of a concern than the resistance of the conductor. This is especially true for RF where frequency components will regularly exceed 1GHz.

Below would be a more accurate representation of how a trace acts in high frequencies than its ideal counterpart.



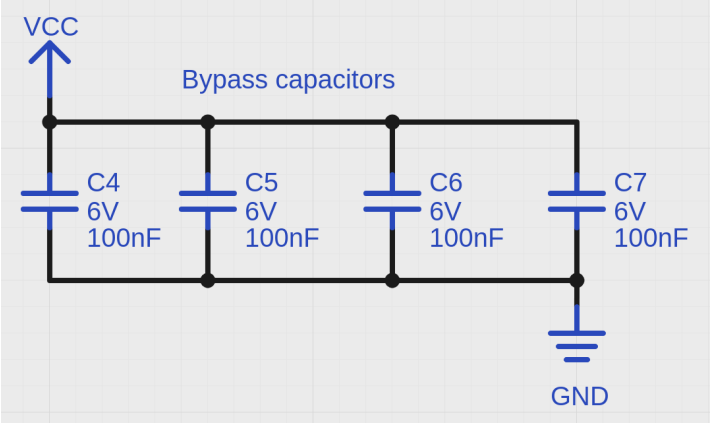
A way to reduce the impact of trace impedance is to keep traces as short as possible. But sometimes it will still be necessary to deliberately choose the trace impedance so that the circuit will work properly.

This text won’t go into the details of how to do that, since it would easily double the size of this document, but I will recommend a useful tool if you need to calculate trace impedances.

https://www.multi-circuit-boards.eu/en/pcb-design-aid/impedance-calculation.html

Bypass/Decoupling capacitor

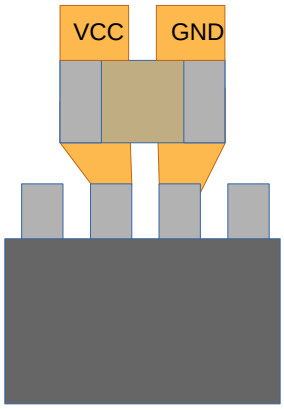
Very often on schematics with ICs you will see a bunch of capacitors connected directly between VCC and GND. If you are not familiar with PCB sedign this might seem odd since for a DC voltage an (ideal) capacitor should act as a break with a DC voltage.



These capacitors fill two main purposes. The first is to filter out AC components of the DC power supply. Since there doesn’t exist an ideal power supply, there will always be some noise on the voltage which can be good to filter out.

Second and often more crucially decoupling capacitors are used as a stabilizer for transient power delivery. Since the conductors going from the power supply to the component has a certain amount of impedance. This means that if a transient of current happens there will be a voltage drop at the component. If there is a capacitor near the device it can act as a small energy store which stabilizes the voltage for transients.

Due to this use these capacitors need to be placed as close to the components power pins as they can so that they can work as efficiently as possible.



Which value of capacitor to use varies depending on the IC but a good starting point for many chips is 100nF. For more demanding components more capacitors in different values might be needed. If you are interested in a deeper understanding on decoupling the following app note by Xilinx is a great resource.

https://www.xilinx.com/support/documentation/application_notes/xapp623.pdf

Manufactures

Over the last few years PCB manufacture has become increasingly available for small scale projects. The lead times on boards have also gone down dramatically.

There is a huge number of manufacturers who has offers for low production number PCBs for less than $10.

At the school we also have a PCB mill which can make a reasonably complex board in half an hour.

But to get from a finished layout you need to generate the correct files to send to the manufacturer.

Gerber

The most common file format for communicating PCB layouts is the **Gerber format**. It's an ASCII based human readable vector format which describes the different layers of a board. The format doesn't contain any information of the schematic or the function of the circuit since this isn't needed for manufacturing the layout.

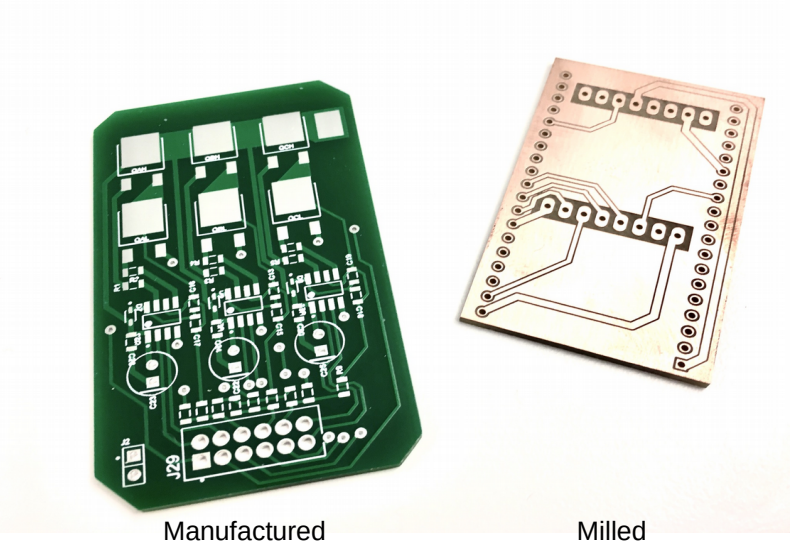
Each layer is typically stored as a separate file, and usually the files are delivered in a **.zip** file.

Due to some technical quirks in the manufacturing process holes are stored in a separate format, **Excellion.**

Your ECAD program will most likely be able to export to gerber and excellion with no issue.

Pros and Cons Using the Mill vs. Manufacturer

Finally here's some pros and cons for when using our mill is a good idea versus ordering online from a manufacurer.



Manufactured Milled

The Mill

- Fast – Many designs can be milled in <30 min
- Cheap – No shipping costs and material costs are low
- Trace width down to 0,3mm
- Smallest distance between conductors 0,25mm
- No solder mask or silkscreen
- Vias must be soldered manually
- Maximum two layer designs
- Good for simple designs and fast iteration

Manufacturer

- Slower – lead times including shipping is at least a week
- Can be cheap but fast shipping adds to the cost
- Easy to make many boards
- Trace width down to ~0.2mm (for cheap boards)
- Solder mask and silkscreen included
- Possible to have more than two layers
- A better choice for more complex designs